

Linearization Theory & Steps

Linearization is the process of finding a linear approximation to a nonlinear system near an equilibrium point.

1. Identify Equilibrium (x_e, u_e)

The system is at equilibrium when $\dot{x} = f(x_e, u_e) = 0$. For the Rod-Mass system, we usually choose: - $x_e = [\theta_e, 0]^T$ - $u_e = \tau_e$ (The torque required to balance the system)

2. Taylor Series Expansion

We define small deviations from equilibrium: - $\tilde{x} = x - x_e$ - $\tilde{u} = u - u_e$

The linear system is defined by the Jacobians:

$$A = \left. \frac{\partial f}{\partial x} \right|_{x_e, u_e}, \quad B = \left. \frac{\partial f}{\partial u} \right|_{x_e, u_e}$$
$$C = \left. \frac{\partial h}{\partial x} \right|_{x_e, u_e}, \quad D = \left. \frac{\partial h}{\partial u} \right|_{x_e, u_e}$$

3. The Linear Model

The resulting linear model is:

$$\dot{\tilde{x}} = A\tilde{x} + B\tilde{u}$$

$$\tilde{y} = C\tilde{x} + D\tilde{u}$$

4. Key Derivatives for Rod-Mass

If $f_2(x, u) = \frac{1}{m\ell^2}(\tau - b\dot{\theta} - mg\ell \cos \theta - k_1\theta - k_2\theta^3)$:

- $\frac{\partial f_2}{\partial \theta} = \frac{1}{m\ell^2}(mg\ell \sin \theta - k_1 - 3k_2\theta^2)$
- $\frac{\partial f_2}{\partial \dot{\theta}} = -\frac{b}{m\ell^2}$
- $\frac{\partial f_2}{\partial \tau} = \frac{1}{m\ell^2}$

5. Implementation Strategy

ALWAYS use the numerical linearization tool (`design_model_tool.py`) first. It is faster and less prone to sign errors. Use the analytical derivatives above only to “sanity check” the numerical results or if the exam explicitly asks for the symbolic form.